

Programming Fundamentals

Loops

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Understanding Loop

Consider the following pattern of output, consisting of a repeated sequence of numbers, like the one below:

For example:

10...9...8...7...6...5...4...3...2...1... Liftoff!

What is a loop?

A loop is a programming structure that repeats a block of statements until a specified condition is satisfied.

How to code the basic C++ for loop

Syntax of a for loop:

```
for (initialization; condition; update)  
{  
    C++ statement(s);  
}
```

```
6      int counterSize = 0;  
7      cout << "Enter the loop counter size: ";  
8      cin >> counterSize;  
9      for (int count = counterSize; count > 0; count--) {  
10         cout << count << "...";  
11     }  
12     cout << " Liftoff!";
```

The output:

10...9...8...7...6...5...4...3...2...1... Liftoff!

Example of while and do-while loops

- While loop (pre-test)

```
6   int counterSize = 0;
7   cout << "Enter the loop counter size: ";
8   cin >> counterSize;
9   while (counterSize > 0) {
10      cout << counterSize-- << "...";
11  }
12  cout << " Liftoff!";
```

- Do-while (post-test)

```
6   int counterSize = 0;
7   cout << "Enter the loop counter size: ";
8   cin >> counterSize;
9   do {
10      cout << counterSize << "...";
11  } while (--counterSize > 0);
12  cout << " Liftoff!";
```

The important point to note, wrong logic can cause infinite loop.

Summary

- Loop allows the user to repeat a block of statements multiple times.
- ***For, while, or do-while*** loops repeatedly execute a block of statements until the conditional expression evaluates to false.
- The increment and decrement operators use to control the number of iteration in the loop.
- The condition expression determines when the loop end.
- Be aware of the danger of an infinite loop.

Questions

